

CLUSTER ANALYSIS

Clustering Algorithms & Validation

Clustering Algorithms

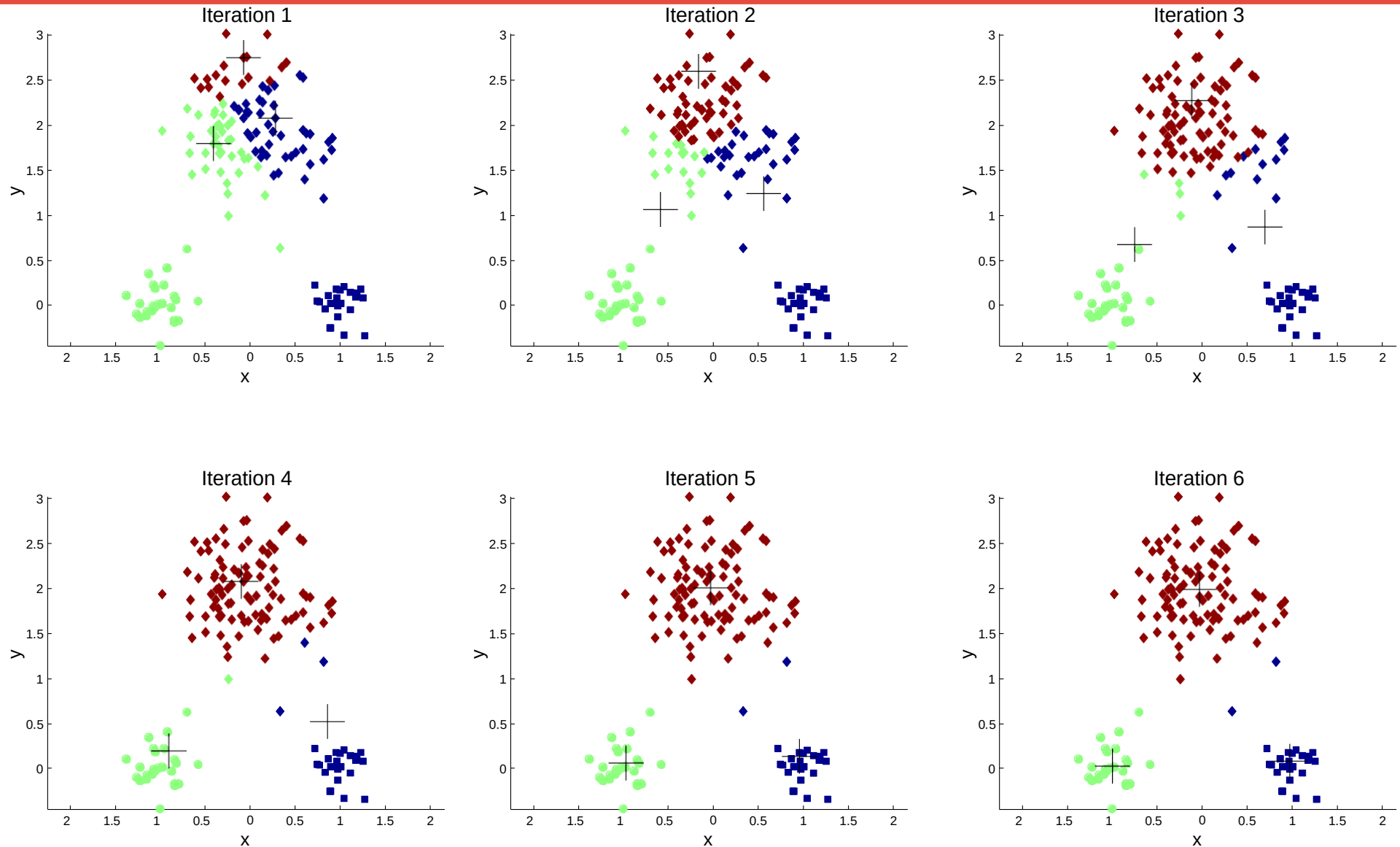
- K-means clustering
- Hierarchical clustering
- Density-based clustering

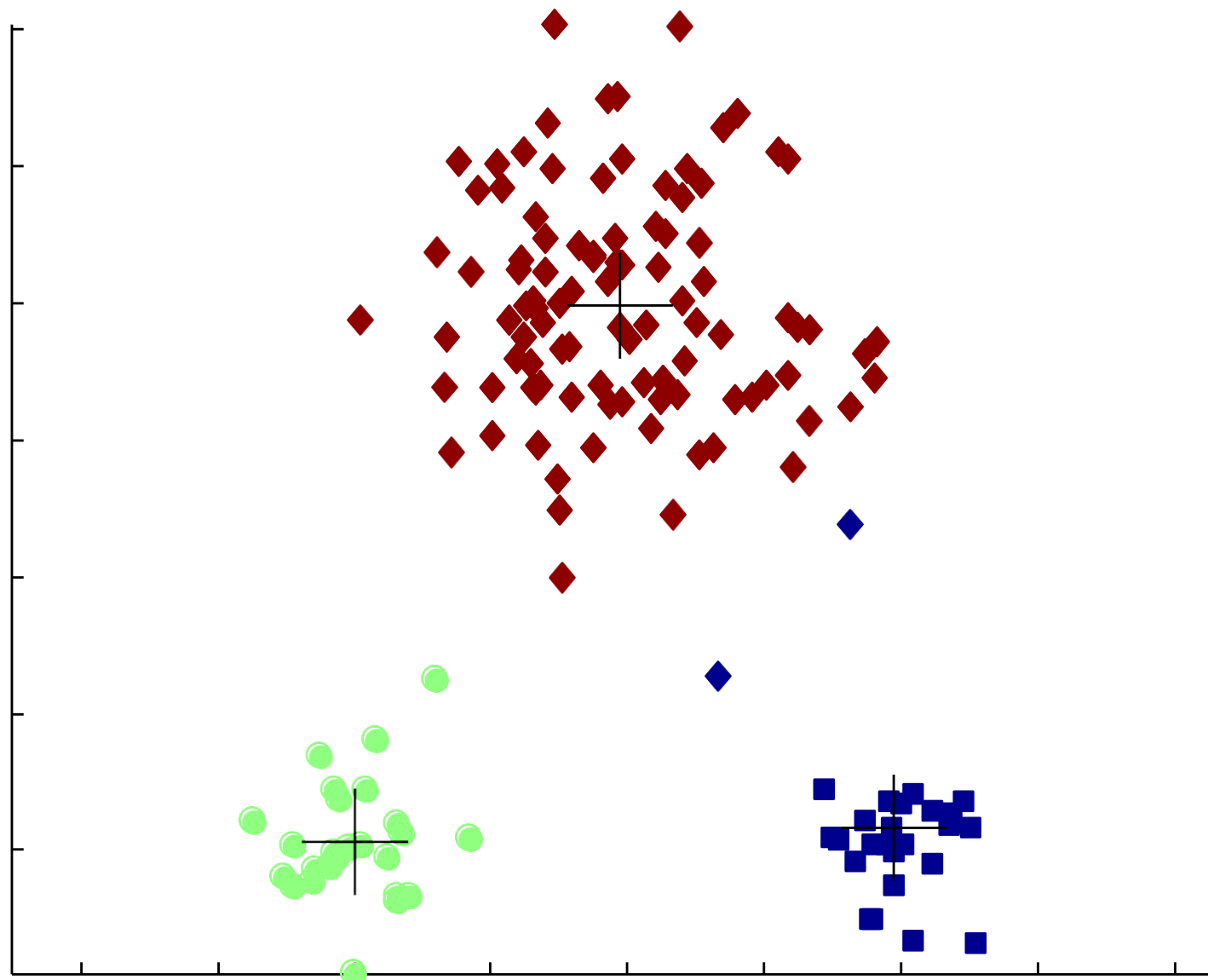
The Basic K-means Algorithm

- Choose K initial centroids, where k is a user-specified parameter, namely, the number of clusters desired
- Each point is then assigned to the closest centroid, and each collection of points assigned to a centroid is a cluster
- Centroid of each cluster is then updated based on the points assigned to the cluster
- Repeat the assignment and update steps until no point changes clusters, or equivalently, until the centroids remain the same

- 
- 1: Select K points as the initial centroids.
 - 2: **repeat**
 - 3: Form K clusters by assigning all points to the closest centroid.
 - 4: Recompute the centroid of each cluster.
 - 5: **until** The centroids don't change

Example of K-means Clustering





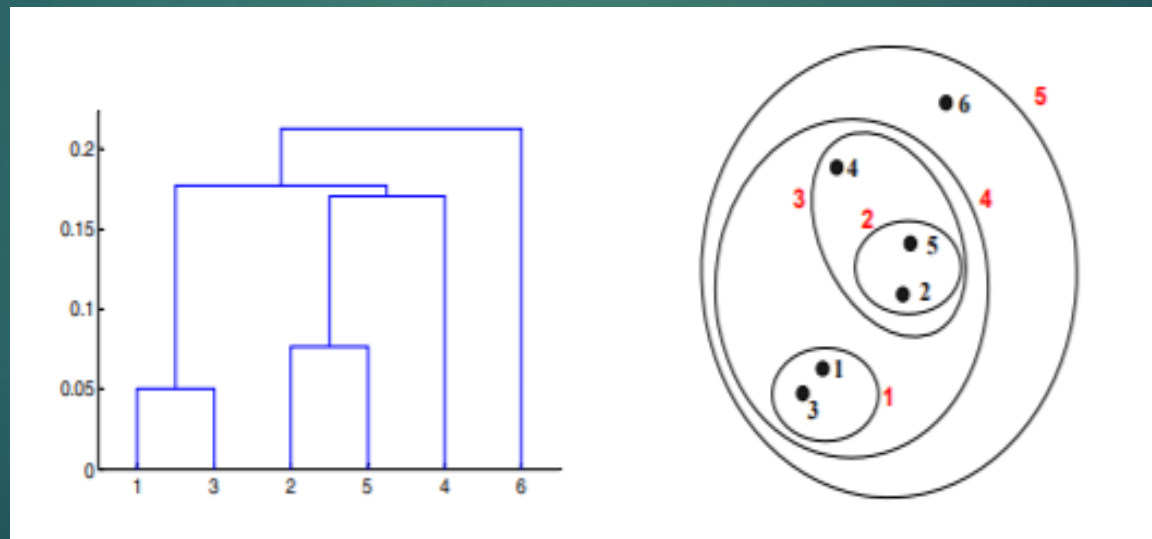
- 
- For some combinations of proximity functions and types of centroids, k means always converges to a solution;
 - K-means reaches a state in which no points are shifting from one cluster to another, and hence, the centroids don't change

Agglomerative Hierarchical Clustering

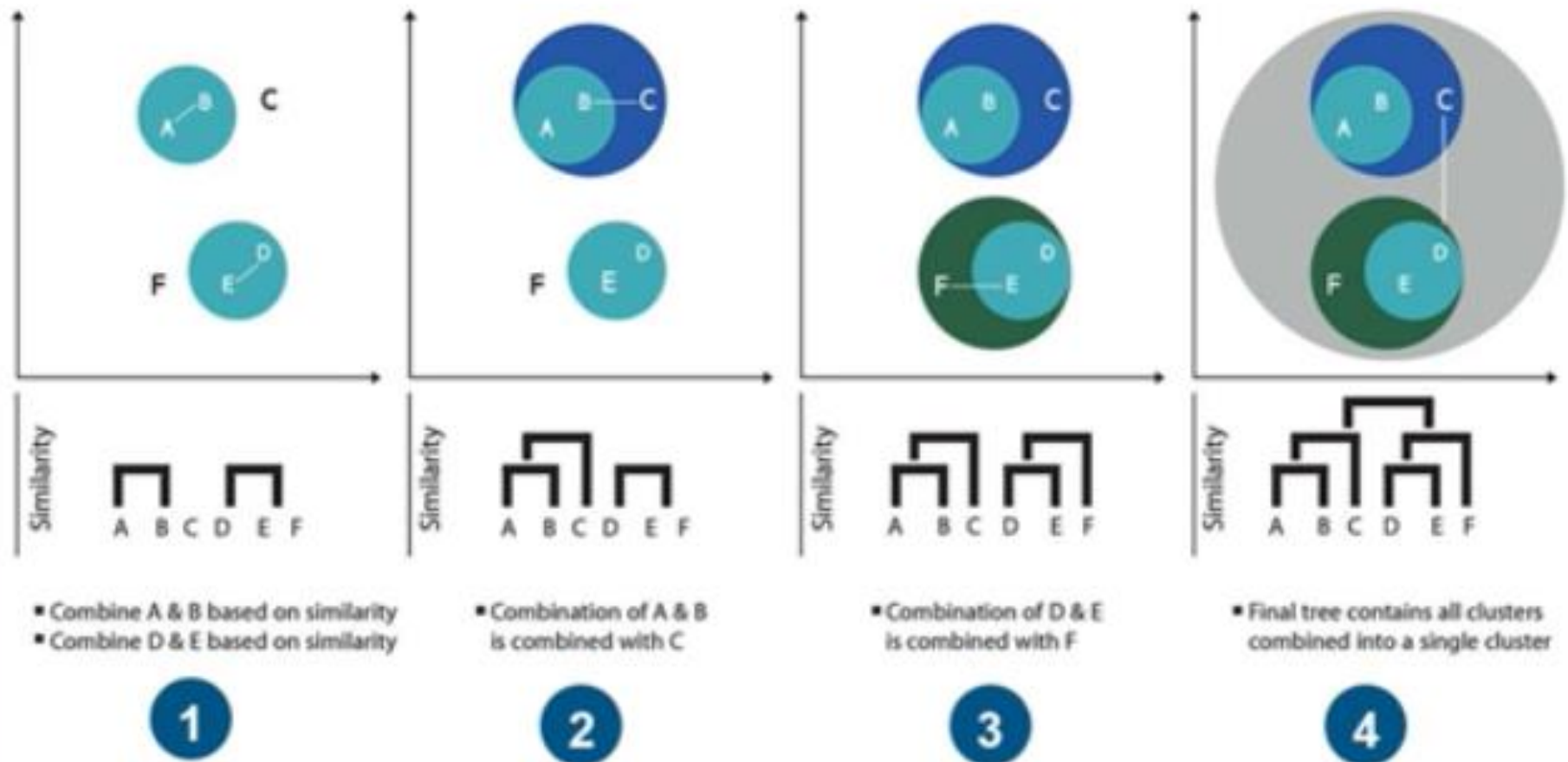
Agglomerative:

- Start with the points as individual clusters
- At each step, merge the closest pair of clusters
- This requires defining a notion of cluster proximity

- A hierarchical clustering is often displayed graphically using a tree-like diagram called a **dendrogram**
- Displays both the cluster-subcluster relationships and the order in which the clusters were merged (agglomerative view)



Hierarchical Clustering



Basic Agglomerative Hierarchical Clustering Algorithm

- Start with individual points as clusters, successively merge the two closest clusters until only one cluster remains

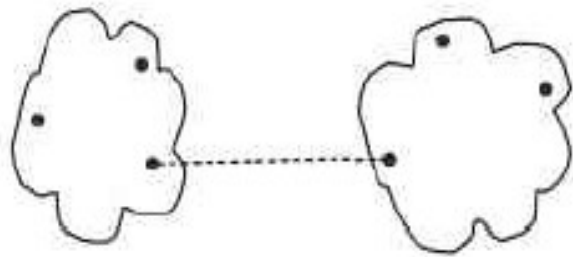
Algorithm 8.3 Basic agglomerative hierarchical clustering algorithm.

- 1: Compute the proximity matrix, if necessary.
- 2: **repeat**
- 3: Merge the closest two clusters.
- 4: Update the proximity matrix to reflect the proximity between the new cluster and the original clusters.
- 5: **until** Only one cluster remains.

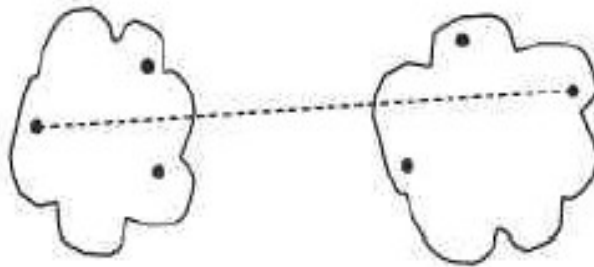
Defining Proximity between Clusters

- Definition of cluster proximity differentiates the various agglomerative hierarchical techniques
- **Graph-based definitions of cluster proximity**
- **MIN** defines cluster proximity as the proximity between the closest two points that are in different clusters,
- Graph terms: the shortest edge between two nodes in different subsets of nodes
- **MAX** takes the proximity between the farthest two points in different clusters to be the cluster proximity,
- Graph terms: the longest edge between two nodes in different subsets of nodes

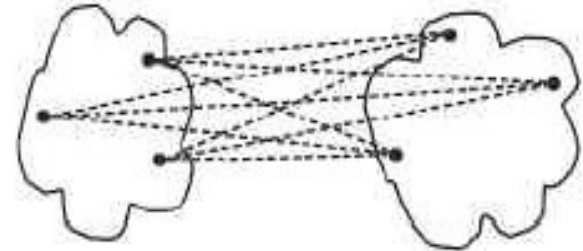
- **Group average** technique defines cluster proximity to be the average pair wise proximities (average length of edges) of all pairs of points from different clusters



(a) MIN (single link.)



(b) MAX (complete link.)



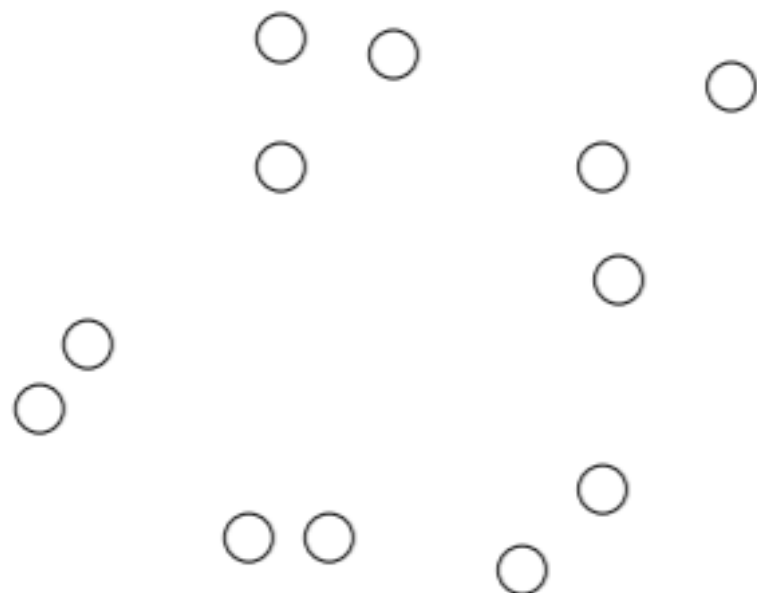
(c) Group average.

Defining Proximity between Clusters

- **Prototype-based view**
- When using centroids, the cluster proximity is commonly defined as the proximity between cluster centroids

Starting Situation

- Start with clusters of individual points and a proximity matrix



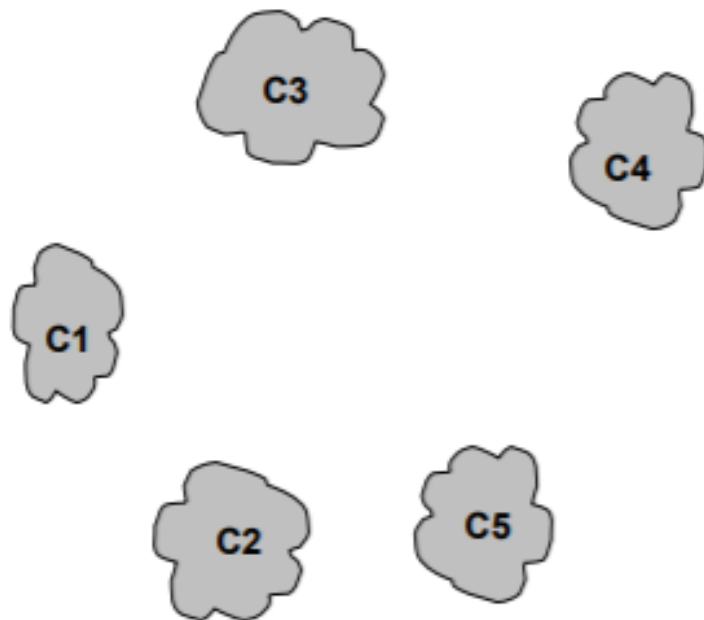
	p1	p2	p3	p4	p5	...
p1						
p2						
p3						
p4						
p5						
.						
.						
.						

Proximity Matrix



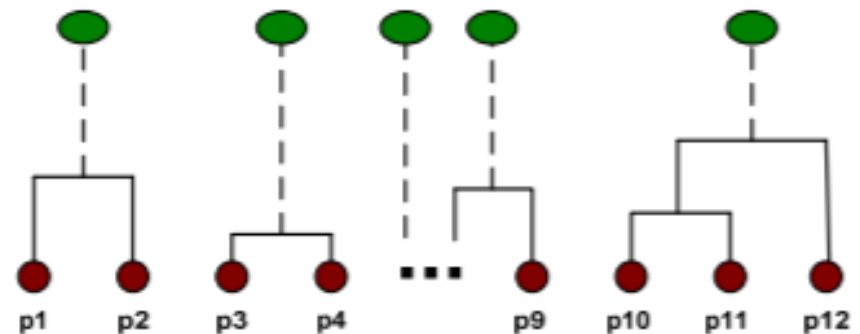
Intermediate Situation

- After some merging steps, we have some clusters



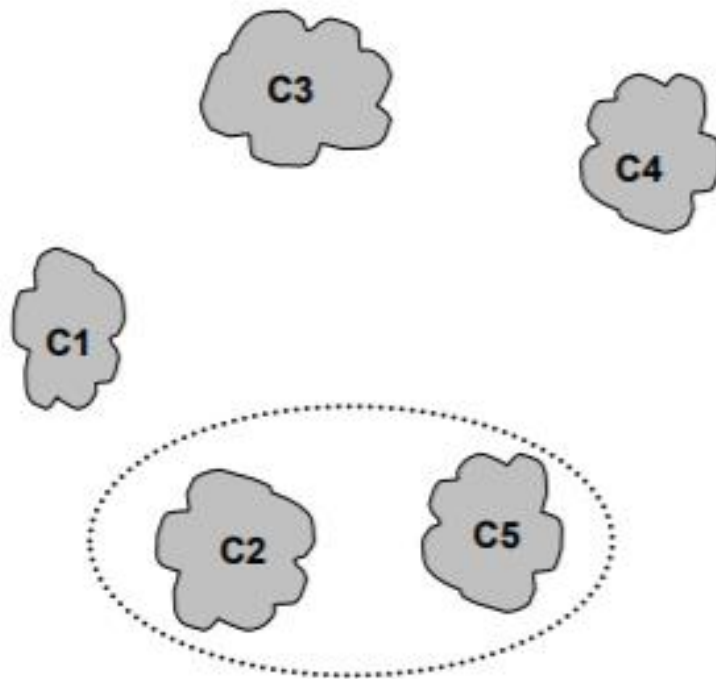
	C1	C2	C3	C4	C5
C1					
C2					
C3					
C4					
C5					

Proximity Matrix



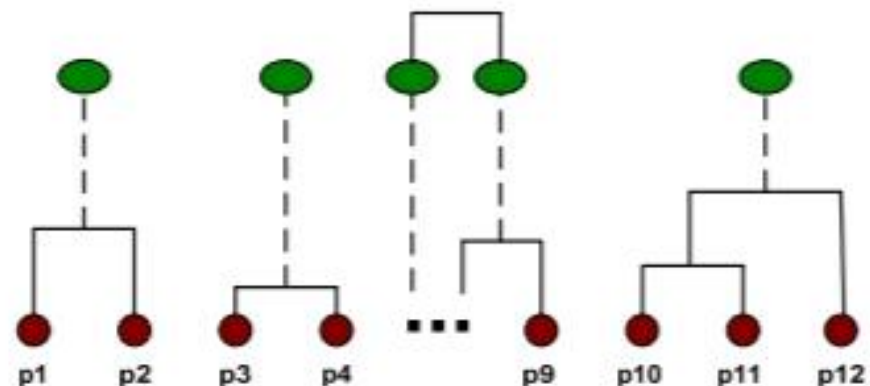
Intermediate Situation

- We want to merge the two closest clusters (C2 and C5) and update the proximity matrix.



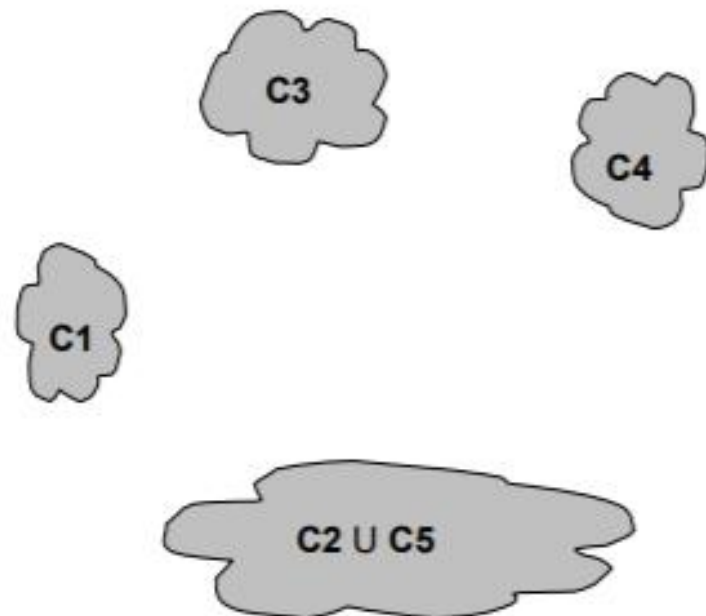
	C1	C2	C3	C4	C5
C1					
C2					
C3					
C4					
C5					

Proximity Matrix



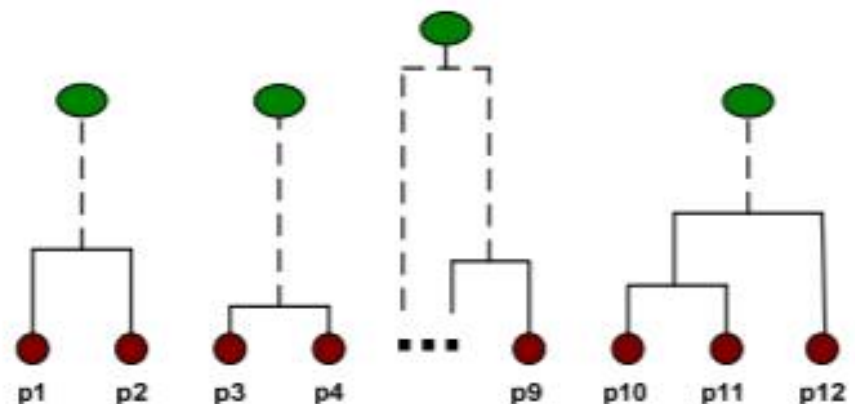
After Merging

- The question is “How do we update the proximity matrix?”

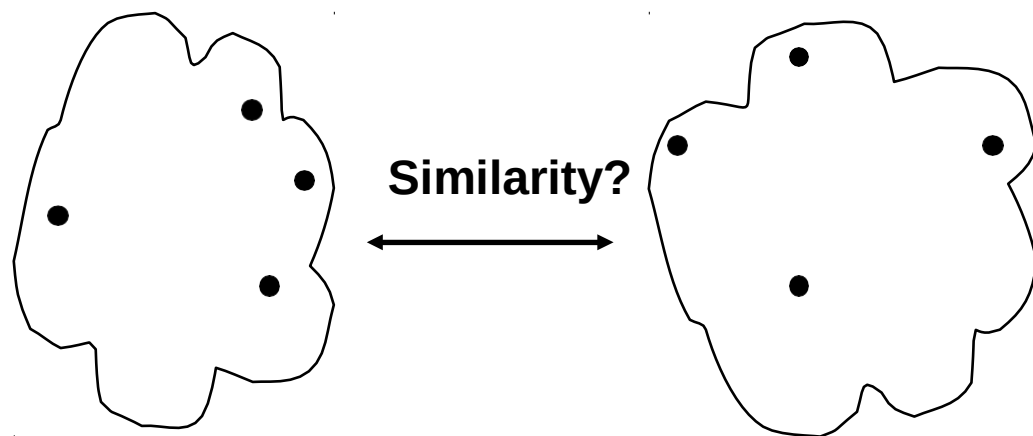


		C1	$C2 \cup C5$	C3	C4
C1			?		
$C2 \cup C5$		?	?	?	?
C3			?		
C4			?		

Proximity Matrix



How to Define Inter-Cluster Distance

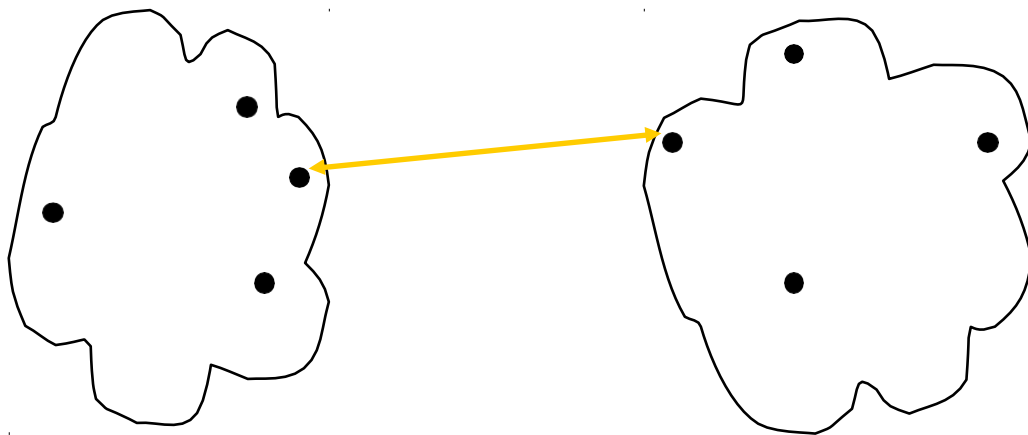


- ☐ MIN
- ☐ MAX
- ☐ Group Average
- ☐ Distance Between Centroids
- ☐ Other methods driven by an objective function
 - Ward's Method uses squared error

	p1	p2	p3	p4	p5	...
p1						
p2						
p3						
p4						
p5						
.						
.						
.						

Proximity Matrix

How to Define Inter-Cluster Similarity

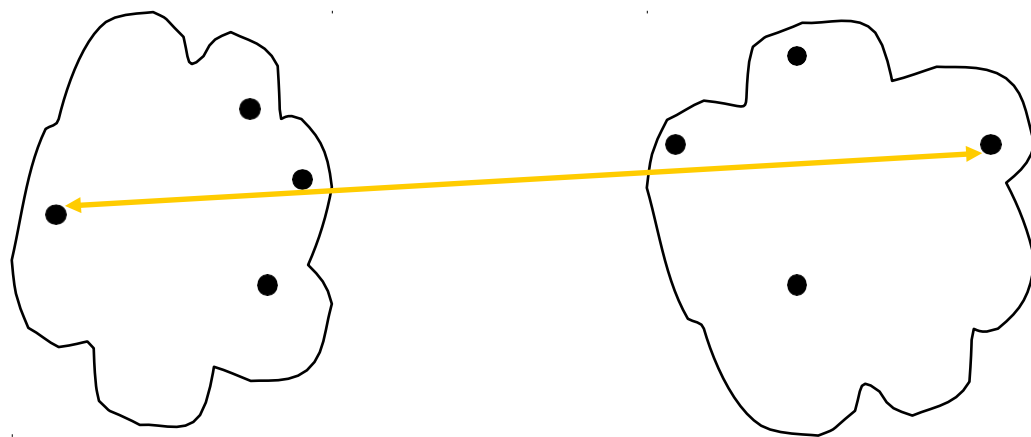


- ☐ MIN
- ☐ MAX
- ☐ Group Average
- ☐ Distance Between Centroids
- ☐ Other methods driven by an objective function
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	p1	p2	p3	p4	p5	...
p1						
p2						
p3						
p4						
p5						
.						
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Proximity Matrix

How to Define Inter-Cluster Similarity

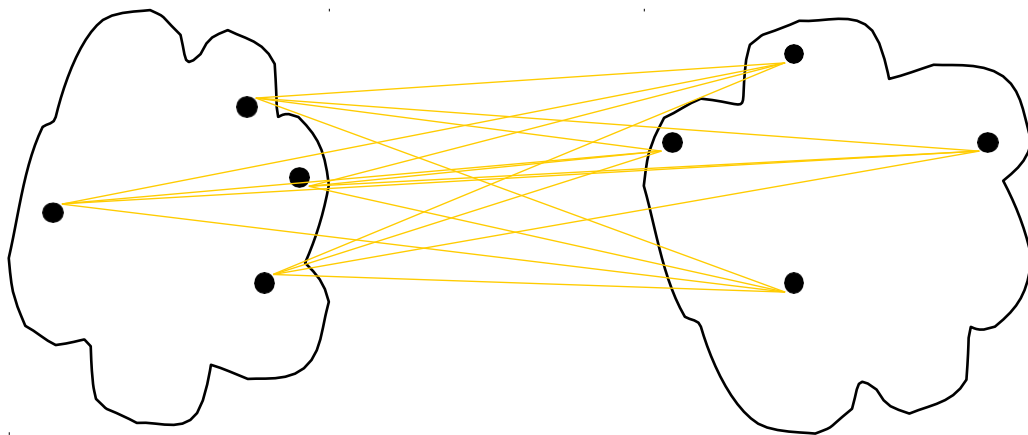


- ☐ MIN
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p1						
p2						
p3						
p4						
p5						
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Proximity Matrix

How to Define Inter-Cluster Similarity

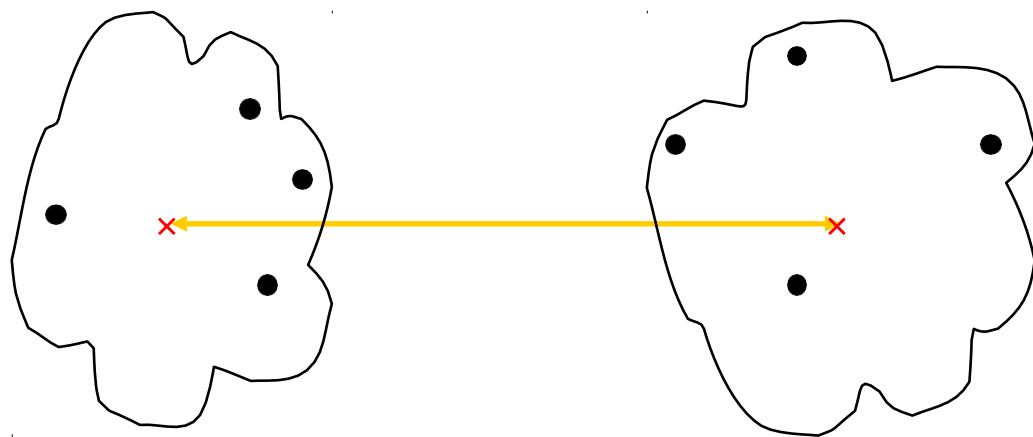


- ☐ MIN
- ☐ MAX
- ☐ Group Average
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	p1	p2	p3	p4	p5	...
p1						
p2						
p3						
p4						
p5						
.						

· **Proximity Matrix**

How to Define Inter-Cluster Similarity



- ☐ MIN
- ☐ MAX
- ☐ Group Average
- ☐ Distance Between Centroids
- ☐ Other methods driven by an objective function
 - Ward's Method uses squared error

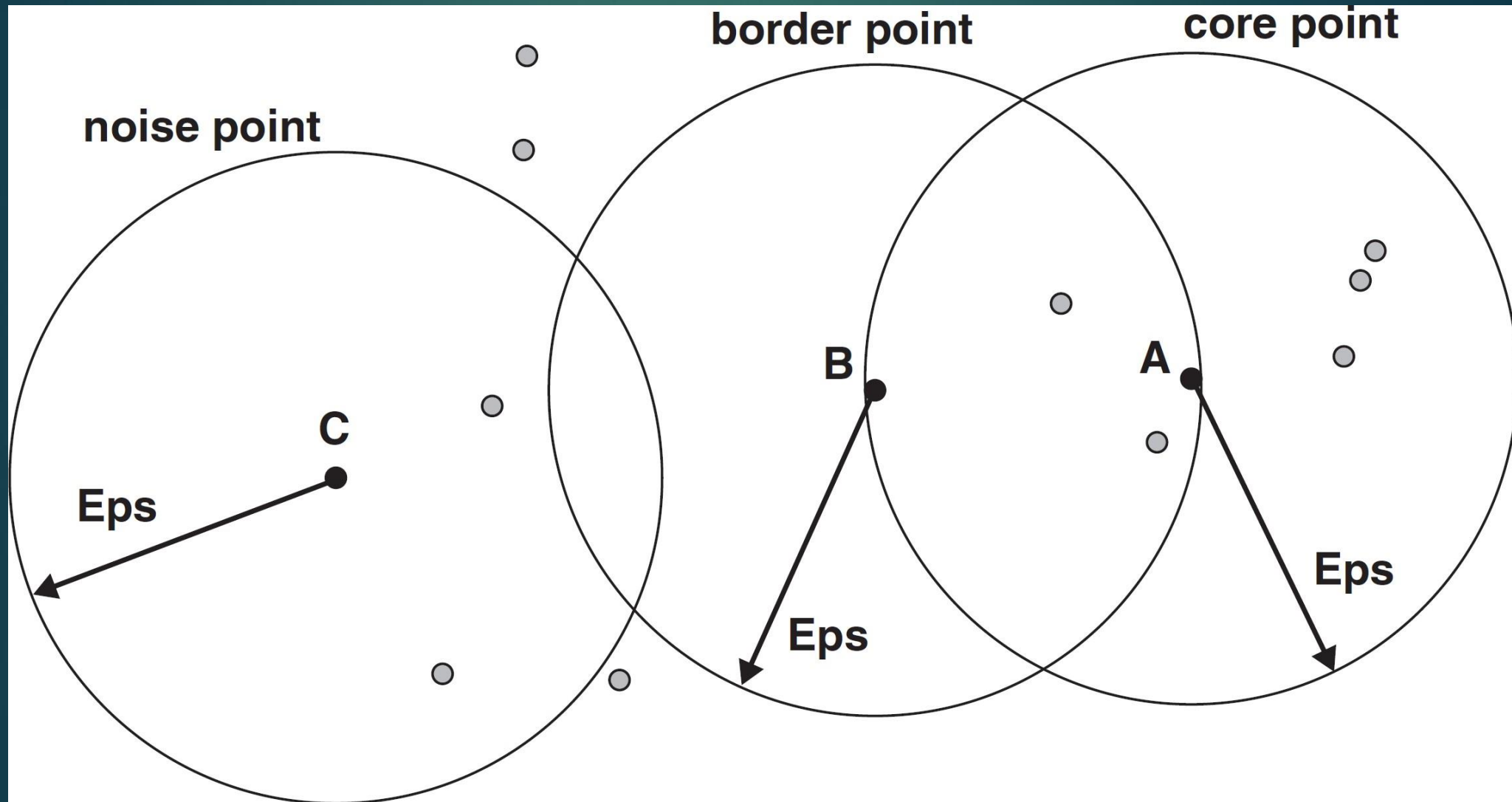
	p1	p2	p3	p4	p5	...
p1						
p2						
p3						
p4						
p5						
.						
.						

• **Proximity Matrix**

DBSCAN (Density-Based Spatial Clustering of Applications with Noise) Algorithm

- DBSCAN is a density-based algorithm
Density = number of points within a specified radius (Eps)
- A point is a **core point** if it has at least a specified number of points (MinPts) within Eps
- These are points that are at the interior of a cluster
Counts the point itself
- A **border point** is not a core point, but is in the neighborhood of a core point
- A **noise point** is any point that is not a core point or a border point

Core, Border, and Noise Points



DBSCAN Algorithm

- Eliminate noise points
- Perform clustering on the remaining points

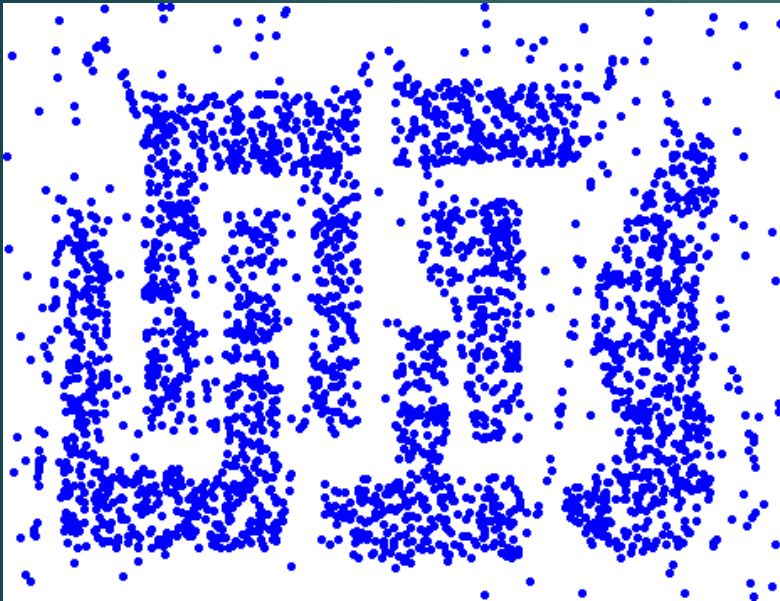
Algorithm 7.5 DBSCAN algorithm.

- 1: Label all points as core, border, or noise points.
 - 2: Eliminate noise points.
 - 3: Put an edge between all core points within a distance Eps of each other.
 - 4: Make each group of connected core points into a separate cluster.
 - 5: Assign each border point to one of the clusters of its associated core points.
-

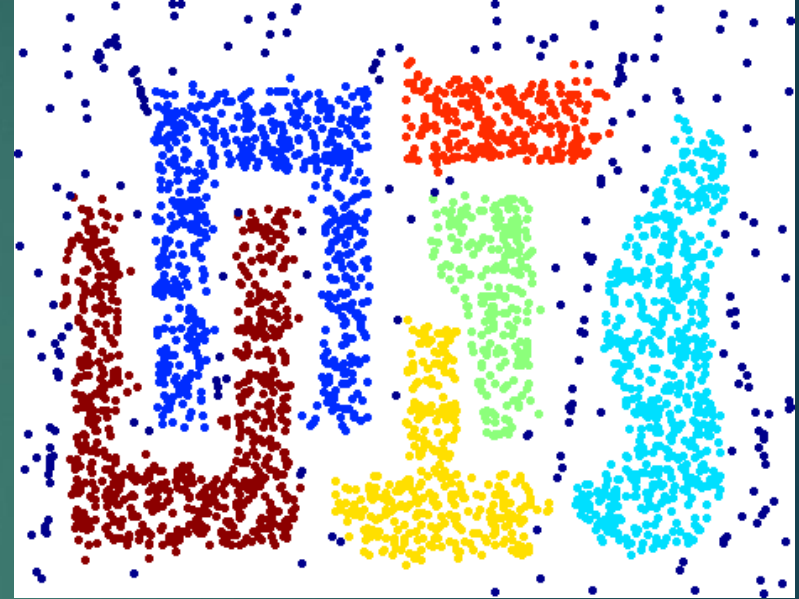
DBSCAN algorithm

1. Label all points as core, border, or noise points
2. Eliminate noise points
3. Put the edge between all core points that are within Eps of each other
4. Make each group of connected core points into a separate cluster
5. Assign each border point to one of the clusters of its associated core points

When DBSCAN Works Well



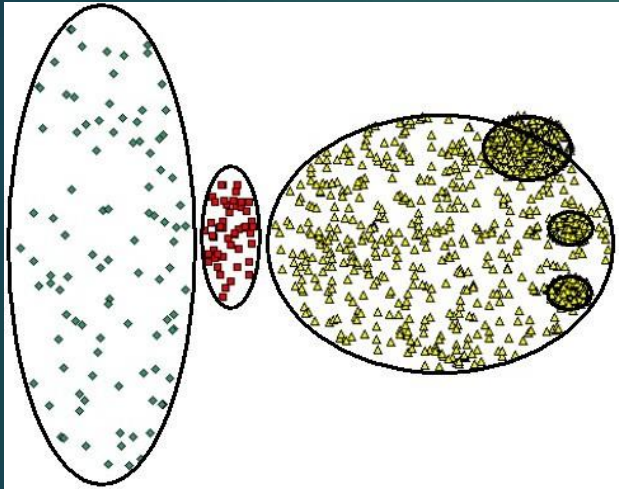
Original Points



Clusters

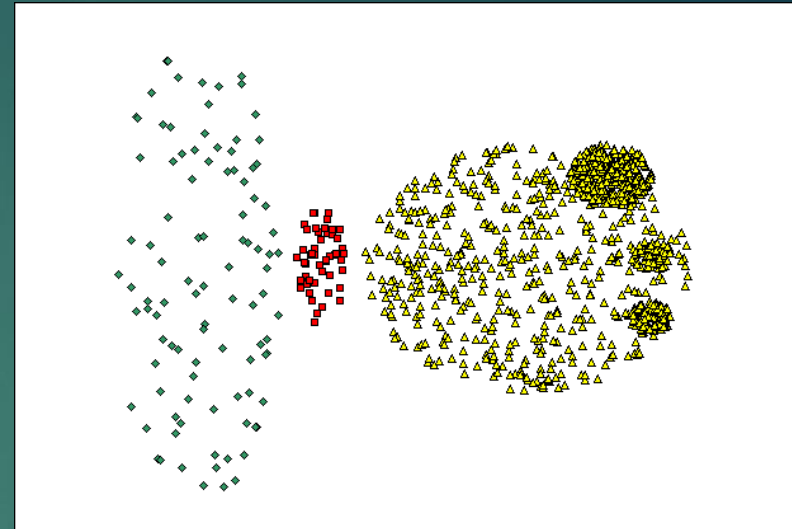
- Resistant to Noise
- Can handle clusters of different shapes and sizes

When DBSCAN Does NOT Work Well

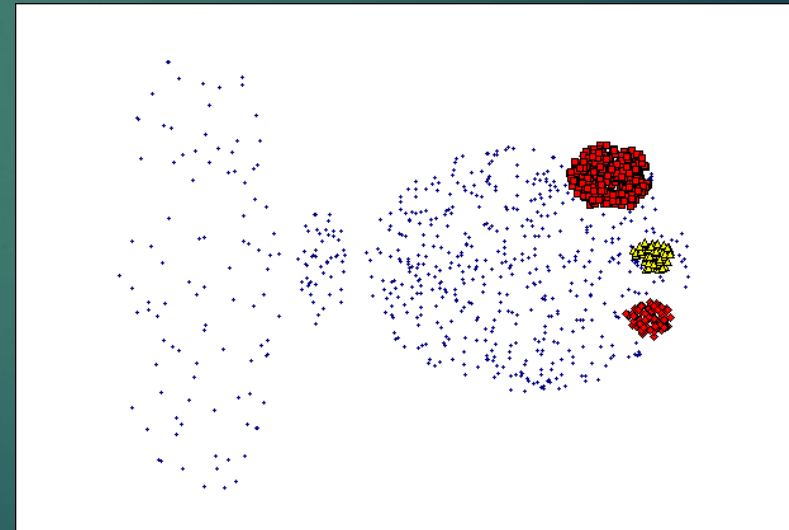


Original Points

- Varying densities
- High-dimensional data



(MinPts=4, Eps=9.75).



(MinPts=4, Eps=9.92)

Cluster Evaluation

Evaluation measures applied to judge various aspects of cluster validity

Unsupervised:

- Measures the goodness of a clustering structure without respect to external information
- e.g.: measures of cluster compactness, isolation etc.



Supervised:

- Measures the extent to which the clustering structure discovered by a clustering algorithm matches some external structure
- External information about data, typically will be in the form of externally derived class labels for the data objects
- Measure the degree of correspondence between the cluster labels and the class labels



Approach 1:

- Use measures from classification, such as entropy, purity, and the F-measure
- These measures evaluate the extent to which a cluster contains objects of a single class

•Approach 2:

- Related to the similarity measures for binary data
- These measure the extent to which two objects that are in the same class are in the same cluster and vice versa

Classification-Oriented Measures of Cluster Validity

Entropy:

- The degree to which each cluster consists of objects of a single class
- For each cluster, the class distribution of the data is calculated first

- For cluster j we compute p_{ij} , the probability that a member of cluster i belongs to class j as $p_{ij} = m_{ij} / m_i$,
 - (m_i is the number of objects in cluster i , and m_{ij} is the number of objects of class j in cluster i)
- Entropy of each cluster i is calculated using formula
$$e_i = - \sum_{j=1}^L p_{ij} \log_2 p_{ij} \quad (L \text{ is the number of classes})$$
- Total entropy for a set of clusters is calculated as the sum of the entropies of each cluster weighted by the size of each cluster
$$e = \sum_{i=1}^K \frac{m_i}{m} e_i$$
- (K is the number of clusters and m is the total number of data points)

Purity:

- Purity of cluster i is
- Overall purity of a clustering :

$$p_i = \max_j p_{ij}$$

$$purity = \sum_{i=1}^K \frac{m_i}{m} p_i$$

Precision:

- The fraction of a cluster that consists of objects of a specified class
- The precision of cluster i with respect to class j is
 $precision(i,j) = p_{ij}$

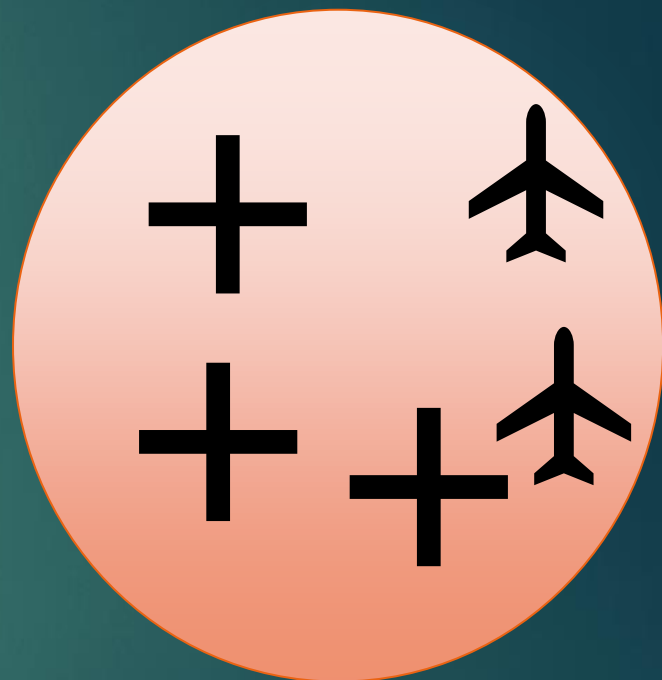
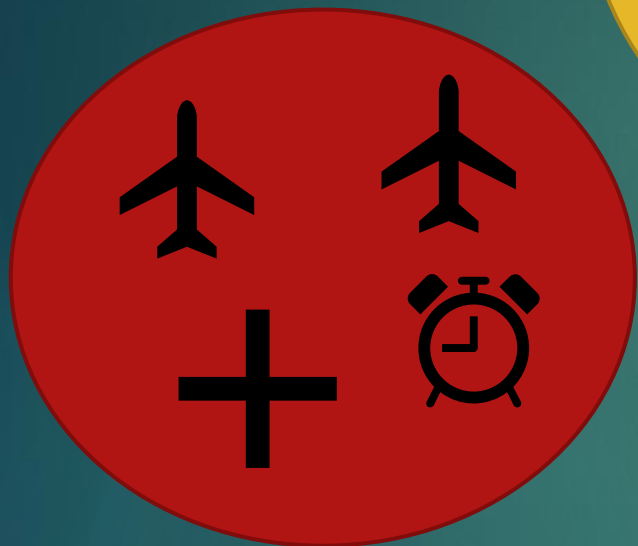
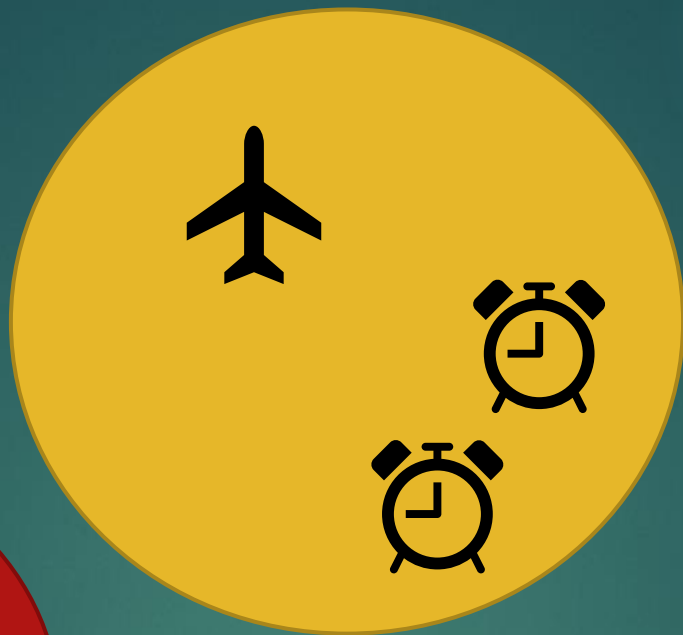
Recall:

- The extent to which a cluster contains all objects of a specified class
- The recall of cluster i with respect to class j is
 $\text{recall}(i,j) = m_{ij} / m_j$, (m_j is the number of objects in class j)

F-measure:

- A combination of both precision and recall that measures the extent to which a cluster contains only objects of a particular class and all objects of that class
- F-measure of cluster i with respect to class j is

$$F(i,j) = (2 \times \text{precision}(i,j) \times \text{recall}(i,j)) / (\text{precision}(i,j) + \text{recall}(i,j))$$



A



B



C

Cluster 1:

Finding P_{ij} :

$$P_{11} = \frac{m_{11}}{m_1}$$

$$P_{11} = \frac{2}{4}$$

$$P_{12} = \frac{m_{12}}{m_1} = \frac{1}{4}$$

$$P_{13} = \frac{m_{13}}{m_1} = \frac{1}{4}$$

Entropy of cluster 1

$$e_1 = -\frac{2}{4} \log\left(\frac{2}{4}\right) - \frac{1}{4} \log\left(\frac{1}{4}\right) - \frac{1}{4} \log\left(\frac{1}{4}\right)$$
$$\approx 1.5$$

Cluster 2:

$$P_{21} = \frac{m_{21}}{m_2} = \frac{1}{3}$$

$$P_{22} = \frac{m_{22}}{m_2} = \frac{2}{3}$$

$$P_{23} = \frac{m_{23}}{m_2} = \frac{0}{3}$$

Entropy of cluster 2:

$$e_2 = -\frac{1}{3} \log\left(\frac{1}{3}\right) - \frac{2}{3} \log\left(\frac{2}{3}\right) - 0 \log\left(\frac{0}{3}\right)$$
$$= 0.9183$$

Cluster 3:

$$P_{31} = \frac{m_{31}}{m_3} = \frac{2}{5}$$

$$P_{32} = \frac{m_{32}}{m_3} = \frac{0}{5}$$

$$P_{33} = \frac{m_{33}}{m_3} = \frac{3}{5}$$

Entropy of cluster 3:

$$e_3 = -\frac{2}{5} \log\left(\frac{2}{5}\right) - \frac{0}{5} \log\left(\frac{0}{5}\right) - \frac{3}{5} \log\left(\frac{3}{5}\right)$$
$$= 0.5710$$

The total entropy = $\frac{4}{12} \times 1.5 + \frac{3}{12} \times 0.9183 + \frac{5}{12} \times 0.5710$

$$\approx \underline{\underline{1.1345}}$$

Purity:

$$\text{Purity of cluster 1: } P_1 = \max\left(\frac{2}{4}, \frac{1}{4}, \frac{1}{4}\right) \\ = \frac{2}{4}$$

$$\text{Purity of cluster 2: } P_2 = \frac{2}{3}$$

$$\text{Purity of cluster 3: } P_3 = \frac{3}{5}$$

$$\begin{aligned} \text{Overall purity of a clustering: } P_{\text{urity}} &= \sum_{i=1}^K \frac{m_i}{m} P_i \\ &= \frac{4}{12} \times \frac{2}{4} + \frac{3}{12} \times \frac{2}{3} + \frac{5}{12} \times \frac{3}{5} \\ &= 0.5833 \end{aligned}$$

Precision:

$$\text{Precision of cluster } i, \text{ w.r.t class } j \text{ precision } (i,j) = P_{ij} = \frac{m_{ij}}{m_i}$$

$$\text{Precision of cluster 1, w.r.t class 1, } P_{11} = \frac{2}{4}$$

Recall:

$$\text{Recall of cluster 1 w.r.t class 1}$$

$$\text{Recall } (1,1) = \frac{m_{11}}{m_1} = \frac{2}{4} ; \text{ Recall } (1,2) = \frac{m_{12}}{m_2} = \frac{4}{1}, \text{ Recall } (1,3) = \frac{4}{1}$$



Cluster	Enter- tainment	Financial	Foreign	Metro	National	Sports	Entropy	Purity
1	3	5	40	506	96	27	1.2270	0.7474
2	4	7	280	29	39	2	1.1472	0.7756
3	1	1	1	7	4	671	0.1813	0.9796
4	10	162	3	119	73	2	1.7487	0.4390
5	331	22	5	70	13	23	1.3976	0.7134
6	5	358	12	212	48	13	1.5523	0.5525
Total	354	555	341	943	273	738	1.1450	0.7203

- Consider cluster 1 and the Metro class
- The precision is $506/677=0.75$, recall is $506/943=0.26$,
- F value is 0.39

Similarity-Oriented Measures of Cluster Validity

- Based on the premise that any two objects that are in the same cluster should be in the same class and vice versa

Involves the comparison of two matrices:

Ideal cluster similarity matrix

- has a 1 in the ij th entry if two objects, i and j are in the same cluster and 0 otherwise

Ideal class similarity matrix

- has a 1 in the ij th entry if two objects i and j belong to the same class, and a 0 otherwise

- Consider five data points p_1, p_2, p_3, p_4, p_5 ;
- two clusters, $C_1 = \{p_1, p_2, p_3\}$ and $C_2 = \{p_4, p_5\}$, and
- two classes $L_1 = \{p_1, p_2\}$ and $L_2 = \{p_3, p_4, p_5\}$

Table 8.10. Ideal cluster similarity matrix.

Point	p1	p2	p3	p4	p5
p1	1	1	1	0	0
p2	1	1	1	0	0
p3	1	1	1	0	0
p4	0	0	0	1	1
p5	0	0	0	1	1

Table 8.11. Ideal class similarity matrix.

Point	p1	p2	p3	p4	p5
p1	1	1	0	0	0
p2	1	1	0	0	0
p3	0	0	1	1	1
p4	0	0	1	1	1
p5	0	0	1	1	1

- We need to compute the following quantities for all pairs of distinct objects
- f_{00} =no of pairs of objects having a different class and a different cluster
- f_{01} =no of pairs of objects having a different class and the same cluster
- f_{10} =no of pairs of objects having the same class and a different cluster
- f_{11} =no of pairs of objects having the same class and the same cluster
- Most frequently used similarity-oriented cluster validity measures:

$$\text{Rand statistic} = \frac{f_{00} + f_{11}}{f_{00} + f_{01} + f_{10} + f_{11}}$$

$$\text{Jaccard coefficient} = \frac{f_{11}}{f_{01} + f_{10} + f_{11}}$$

- 
- $f_{00}=4$
 - $f_{01}=2$
 - $f_{10}=2$
 - $f_{11}=2$
 - Rand statistic = $(2 + 4) / 10 = 0.6$
 - Jaccard coefficient = $2 / (2 + 2 + 2) = 0.33$